Reads over Atom Spaces: A Mathematical View of Context, Retrieval, and Parameter Routing in Language Models

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Abstract

Attention, external retrieval, and parameter-expert routing all compute query-conditioned weighted reads. We provide a common analytical lens based on valued atom spaces $(\Omega, \mathcal{F}, \nu, v)$ and the read operator $R(x;\Omega) = \int_\Omega v(\omega)\, d\mu_x(\omega)$, which reduces to a softmax-weighted value combination under a counting measure. Operations on the reference measure ν express several mechanisms in one place. A *null atom* yields abstention, and the hard activation threshold of gated retrieval is its zero-temperature limit. A matched *anti-atom* is exactly deletion plus a null mass at the deleted key. A query-side margin certifies, at insertion time and without a Lipschitz premise, a radius within which a record remains the top-1 read. An exact softmax-tail bound gives an error-budgeted truncation rule that selects 52–86% of a synthetic corpus at $N = 25{,}000$, so it is a certificate rather than a sublinear method. On CounterFact-500, held-out calibration of a frozen-MiniLM activation threshold raises paraphrase activation from 38.2% to 53.2% at a false-activation cost of 9.6% to 16.1%. The remaining gap to 91% top-1 recall consists of correctly ranked, low-similarity paraphrases: a gate on the rank margin reaches 70.8% activation at 7.0% false activation. Under one scorer, the gated read, a rank-one parametric reference, and GRACE occupy different points of a generalization–locality trade-off set by their key spaces. Tri-space routing with null atoms and first-token steering are reported as diagnostics. We offer a shared vocabulary and empirical calibrations of what retrieval, margin, and tail arguments can certify, not a new editing method.

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